

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a013eb8a77cb5dd05.jsonl`  
- SHA-256: `d87ac3f5d4c52ae88eaad17a0dab1dd505e39bdaefab24ecb28b092e3989cfd5`  
- Source modified: `2026-06-16T03:19:50+00:00`  
- Imported at: `2026-07-05T16:48:26+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:19:04.325Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://arxiv.org/abs/2007.06866>

****Title:**** Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF, WACV 2021)

****Found via:**** Single-stage TAL actionness + boundary heads, small data search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
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If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:19:06.493Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-16T03:19:21.658Z)

Key Claims from arXiv:2007.06866

****Paper Details:****

- Title: "Alleviating Over-segmentation Errors by Detecting Action Boundaries"
- Submitted: July 14, 2020
- Status: Under review

- arXiv ID: 2007.06866 (cs.CV)

****Main Contribution:****

The authors propose the Action Segment Refinement Framework (ASRF), which addresses over-segmentation in temporal action segmentation through a dual-branch architecture. One branch performs frame-wise action classification while the other regresses action boundary probabilities to refine predictions.

****Architecture:****

The framework consists of "a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB)." The ASB classifies frames while the BRB "regresses the action boundary probabilities."

****Reported Performance Improvements:****

The abstract claims their approach delivers improvements of "up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score" over prior state-of-the-art methods.

****Datasets:****

The paper indicates evaluation on "three challenging datasets," though specific dataset names (50Salads, GTEA, Breakfast) are not mentioned in the provided page content.

****Additional Contributions:****

The authors propose a specialized loss function for smoothing action probability transitions across temporal boundaries.

4. assistant (2026-06-16T03:19:24.124Z)

The abstract gives the headline numbers but truncates dataset names and per-metric breakdowns. Let me confirm the dataset names and additional detail to strengthen the claims.

5. user (2026-06-16T03:19:30.781Z)

Analysis of arXiv Paper 2007.06866

Datasets

The abstract mentions evaluation on "three challenging datasets" with specific performance improvements (13.7% in segmental edit distance, 16.1% in segmental F1 score), but ****the specific dataset names are not listed**** in this page excerpt. The datasets 50Salads, GTEA, and Breakfast you referenced are not mentioned here.

Code Availability

The paper states: ****"Our code will be publicly available soon."**** However, no specific license information or repository link is provided on this page. No open source license is specified.

Boundary Refinement Mechanism

The document describes the approach as: ****"The action boundaries predicted by the BRB refine the output from the ASB, which results in a significant performance improvement."**** However, the specific technical details of how this refinement occurs (whether through post-processing, fusion, or another method) are not elaborated in this abstract excerpt.

MS-TCN Comparison

MS-TCN is not mentioned anywhere on this page. No comparative analysis details are provided in the available content.

****Note:**** The full paper PDF would contain these implementation details, dataset specifications, and comparative results not visible in this abstract page.

6. assistant (2026-06-16T03:19:39.050Z)

I have enough from the abstract to extract verifiable claims. The arXiv abstract page is the primary source (the paper itself, by Ishikawa et al.). The dataset names (50Salads, GTEA, Breakfast) and MS-TCN comparison aren't in the abstract excerpt, so I'll flag those as not verified from this page and only assert what's quotable.

7. user (2026-06-16T03:19:42.428Z)

Structured output provided successfully

8. assistant (2026-06-16T03:19:50.422Z)

Extracted claims from the ASRF paper (arXiv:2007.06866, "Alleviating Over-segmentation Errors by Detecting Action Boundaries," Ishikawa et al., submitted 2020-07-14, later WACV 2021).

Source quality: primary (the paper's own arXiv abstract page).

Key findings relevant to the research question (Dimension 1, over-segmentation control):

- ASRF's central mechanism ? a dual-branch design where a Boundary Regression Branch (BRB) explicitly detects action boundaries and refines the Action Segmentation Branch (ASB) output ? directly maps onto the project's diagnosed failure (no inactivity/boundary split, conflating "shuttle moving" with "rally in progress").
- The headline gains are on exactly the over-segmentation-sensitive metrics the project already uses: up to 13.7% segmental edit distance and up to 16.1% segmental F1.
- A boundary-smoothing loss complements the boundary head.

Caveats flagged as unverified from this page:

- Dataset names (50Salads/GTEA/Breakfast) and the MS-TCN baseline comparison are standard for this line of work but were NOT present in the abstract excerpt ? not asserted as verified.
- Code license is not stated ("code will be publicly available soon"), so MIT/Apache/BSD cleanliness cannot be confirmed from this source and needs separate verification against the project's hard licensing constraint.

Structured output has been submitted.